

Introduction to Quantum Computation

High Performance and Disruptive Computing in Remote Sensing School 2026

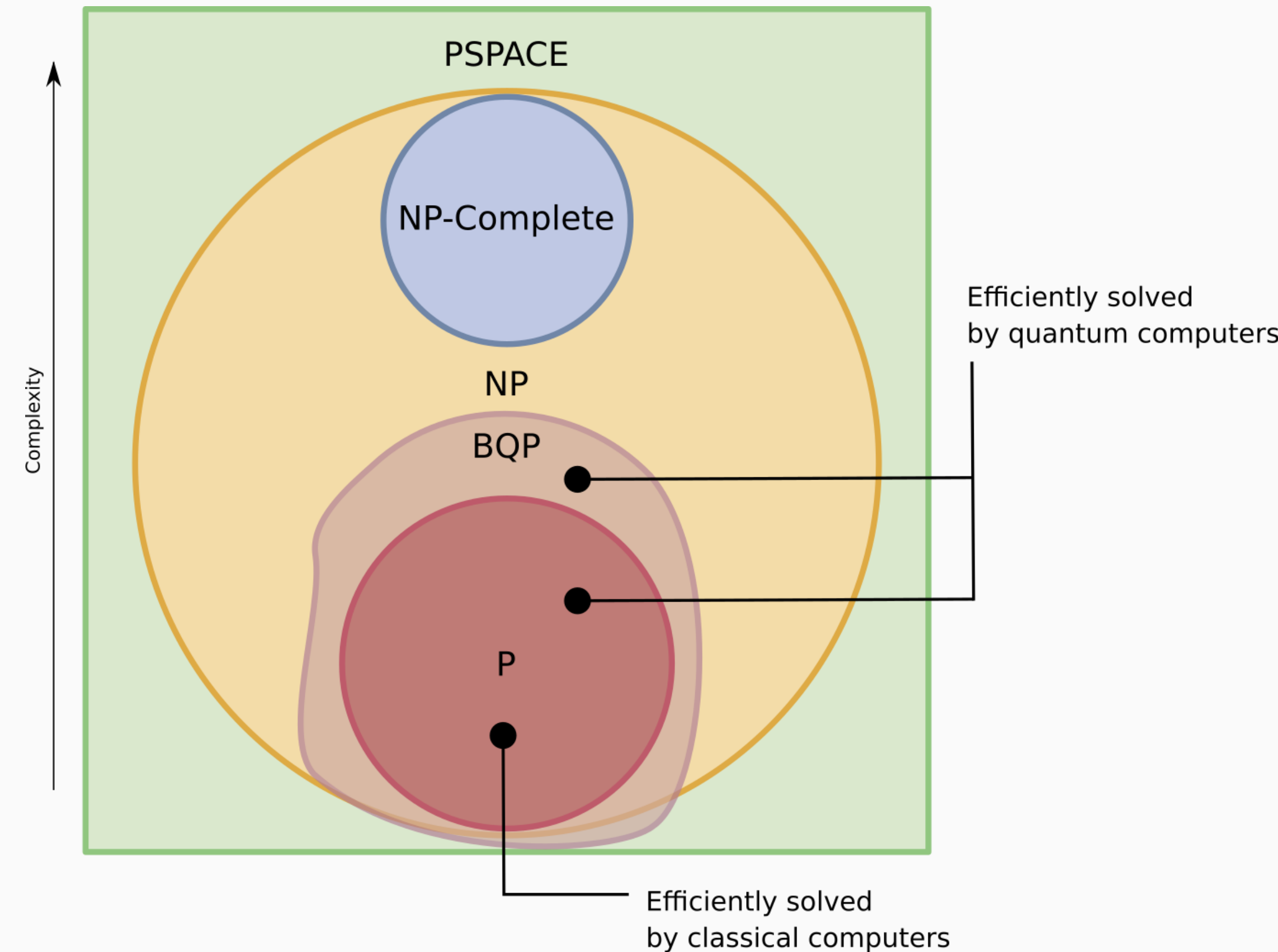
We strongly believe in truly transformative ideas and in the power of compelling partnerships to accelerate the Earth Observation future

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Why quantum computing?

- A **computational complexity class** is a set of decision problems grouped according to how the required computational resources scale asymptotically with input size under a specified model of computation.
- The key point is the **scaling behavior** (e.g., polynomial, exponential, logarithmic), not the exact number of steps.
- **BQP** (Bounded-error Quantum Polynomial time) contains all the problems in P, but may also contain problems in NP!

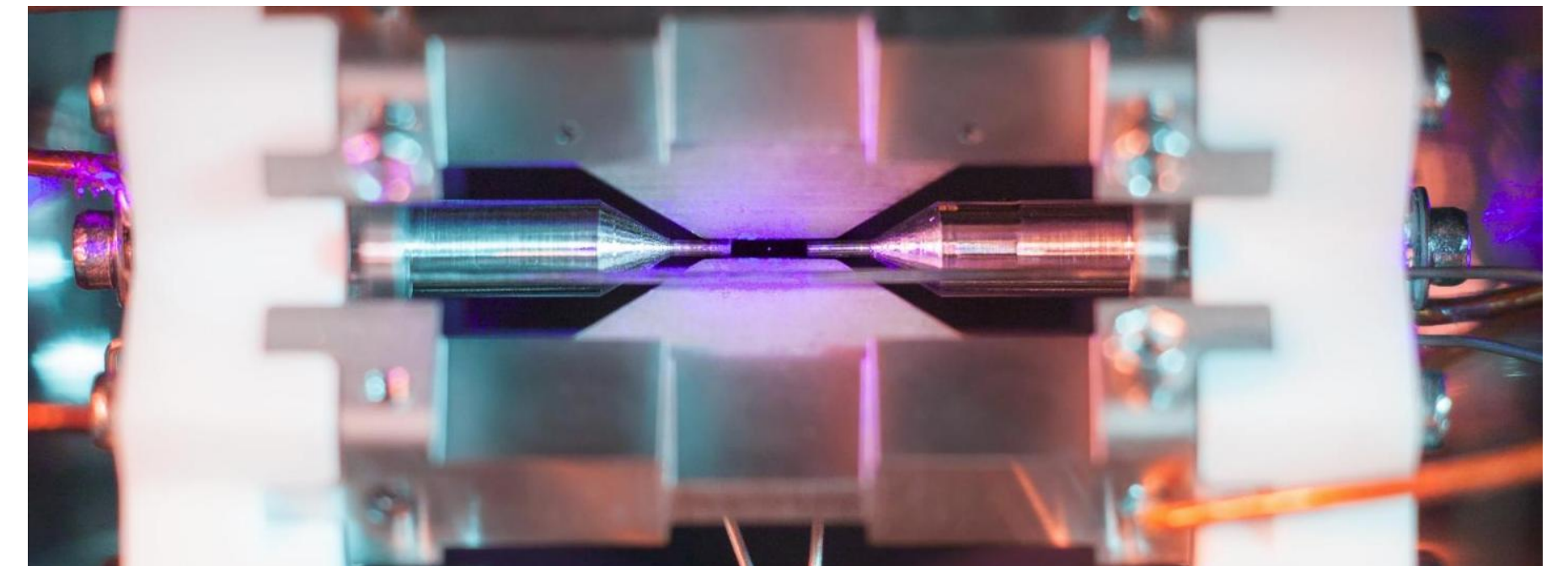
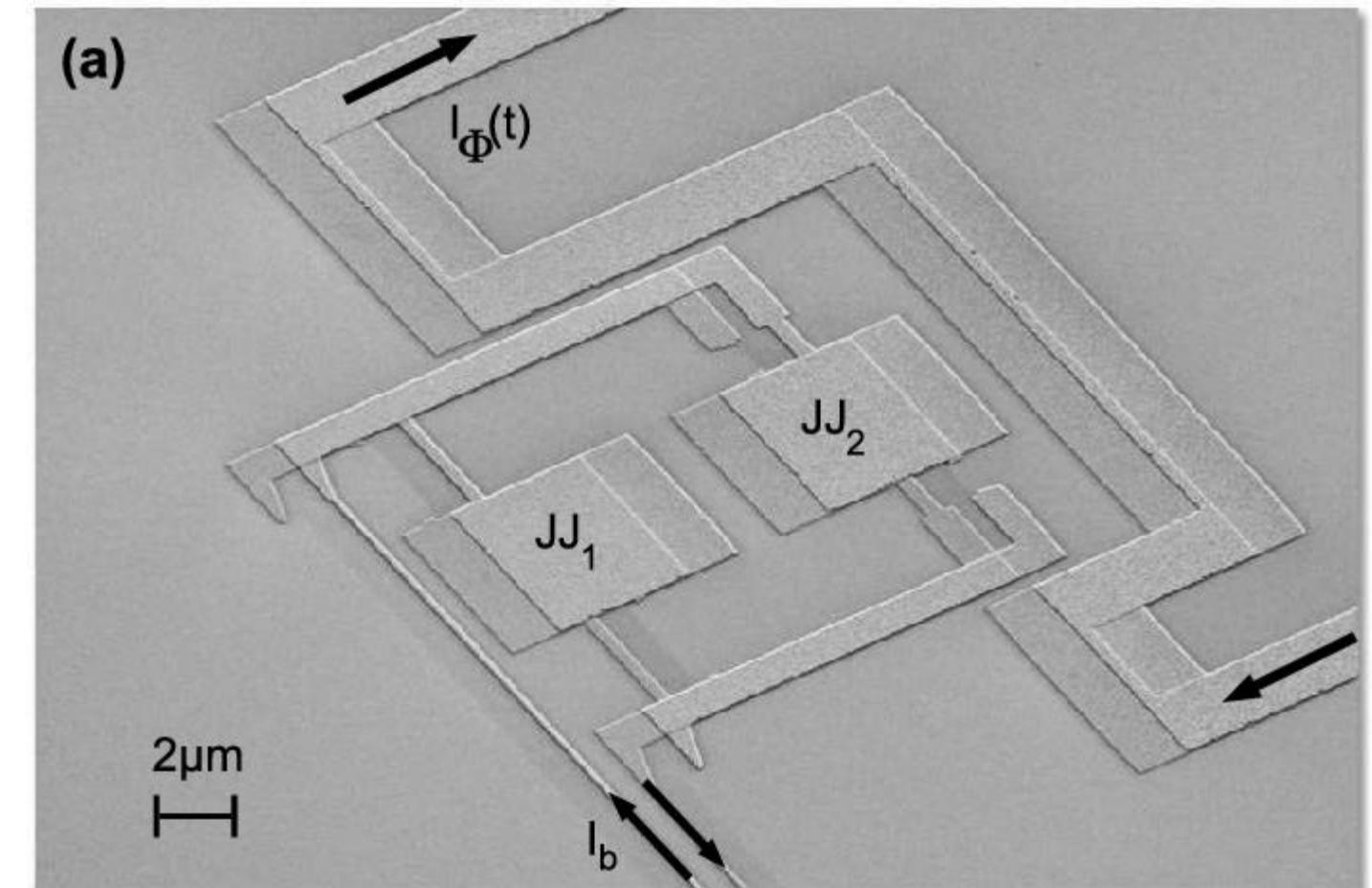


“Information is Physical”

Rolf Landauer, 1991



- All information is inevitably tied to a physical substrate
- Different computing technologies can have different information processing capabilities
- Quantum computing is computational technology, that utilizes laws of quantum mechanics at the tiniest scale



Quantum Computers are “slow”

Otgonbaatar, Soronzonbold, et al. "Quantum computing for climate change detection, climate modeling, and climate digital twin." (2023).

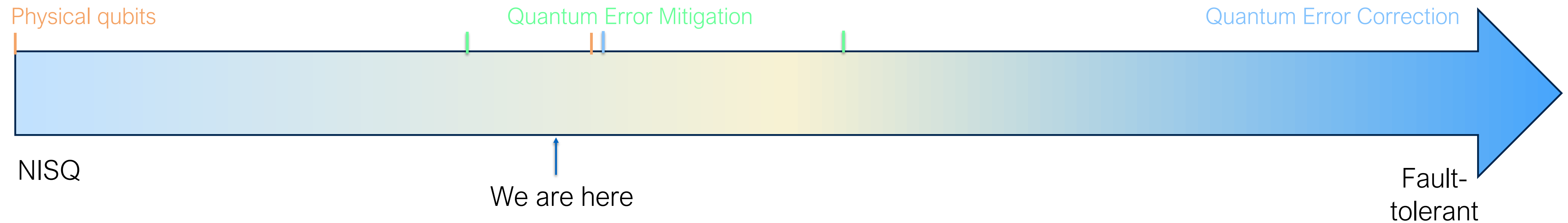
Parameters	SC	T.ions	Photonic	N.atoms	S.spin	NV	CPUs
Clock cycle	1MHz	1KHz	10Hz	1MHz	0.76MHz	1MHz	3GHz
Measurement	660ns	300μs	x	200ms	1.3μs	x	x
2-qubit gate	34ns	200μs	x	< 100μs	x	700ns	x
1-qubit gate	25ns	15μs	x	x	x	9ns	x
Readout fidelity	99.4%	97.3%	50.0%	99.1%	99%	98%	x
1Q fidelity	99.99%	99.99%	99.84%	99.83%	99.99%	99.99%	x
2Q fidelity	99.97%	99.9%	99.69%	99.4%	99.5%	99.2%	x

TABLE I. Resources required to achieve quantum advantage assuming speedups of various polynomial degrees d . We make this comparison against an adversary using distributed classical computing resources that achieve a speedup factor S and report the number of algorithm steps M and the total runtime T^* before a quantum speedup is possible. We make this comparison for both the informal resource “lower bound” we argue for in the main text (using $t_Q \geq 17$ ms and $t_C \leq 33$ ns), and for the specific example of quantum simulated annealing applied to the Sherrington-Kirkpatrick model using the quantum and classical implementations discussed in Refs. [12,49] (giving $t_Q = 440$ ms and $t_C = 7$ ns).

Polynomial degree	Parallelism speedup S	Resource “lower bound”		Simulated annealing	
		Iterations M	Runtime T^*	Iterations M	Runtime T^*
Quadratic, $d = 2$	1	5.2×10^5	2.4 h	6.3×10^7	320 days
	10^3	5.2×10^8	100 days	6.3×10^{10}	880 years
	10^6	5.2×10^{11}	280 years	6.3×10^{13}	880 millennia
Cubic, $d = 3$	1	7.2×10^2	12 s	7.9×10^3	58 min
	10^3	2.3×10^4	6.4 min	2.5×10^5	1.3 days
	10^6	7.2×10^5	3.4 h	7.9×10^6	40 days
Quartic, $d = 4$	1	8.0×10^1	1.4 s	4.0×10^2	2.9 min
	10^3	8.0×10^2	14 s	4.0×10^3	29 min
	10^6	8.0×10^3	2.3 min	4.0×10^5	4.9 h

Babbush, Ryan, et al. "Focus beyond quadratic speedups for error-corrected quantum advantage." *PRX quantum* (2021)

Quantum Computing Development Timeline



Noisy Intermediate-Scale Quantum

- Physical qubits, not logical
- Around 50-1000 of qubits
 - Error acumulates
 - Short routines
- Heuristic and benchmarking algorithms

Quantum Error Mitigation

- Transitive era techniques
 - $\uparrow$ time $\downarrow$ errors
- Useful both $\leftarrow$ and $\rightarrow$

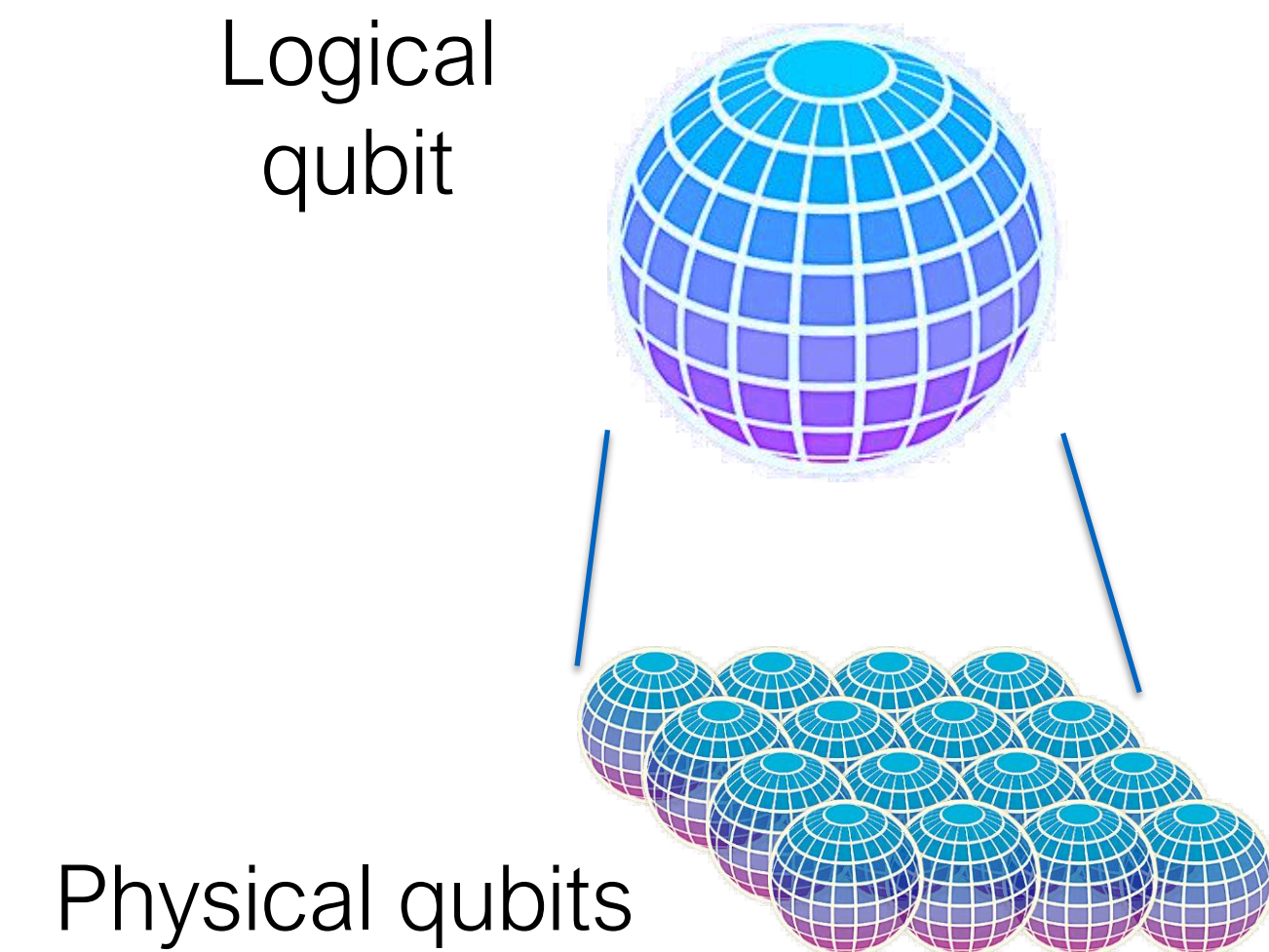
Fault tolerant

- Physical qubits below error threshold
 - Error correction codes
- 1 logical qubit = 10-1000 physical qubits
- Powerful algorithms

- Size matters in robust quantum advantage
- In physical qubits errors accumulate, prohibiting complicated quantum routines (Shor's, Grover's algorithms)
- Error correction is necessary!
- Representing single logical qubit with many error-corrected physical qubits

Threshold theorems:

“Below some error threshold for physical qubits, adding more of them decreases the error in logical qubit”



For Shor's algorithm: $\sim 10^3$ logical qubits $\longrightarrow$ $\sim 10^6$ physical qubits!

Will we achieve full quantum error correction?
With the current steady progress we should be optimistic!

Noisy

- Quantum devices are exposed to noise (thermal fluctuations, environment, imperfections of gates,...), error rate $10^{-3} - 10^{-5}$.
- This limits the complexity and accuracy of the computations they can perform.
- Above error thresholds – no error-correction

Intermediate-Scale

- $50 - 10^3$ physical qubits
- 50 qubits milestone, $2^{50} \approx 10^{15}$
- Beyond what can be simulated by brute force using the most powerful existing digital supercomputers.
- Not enough for complex algorithms

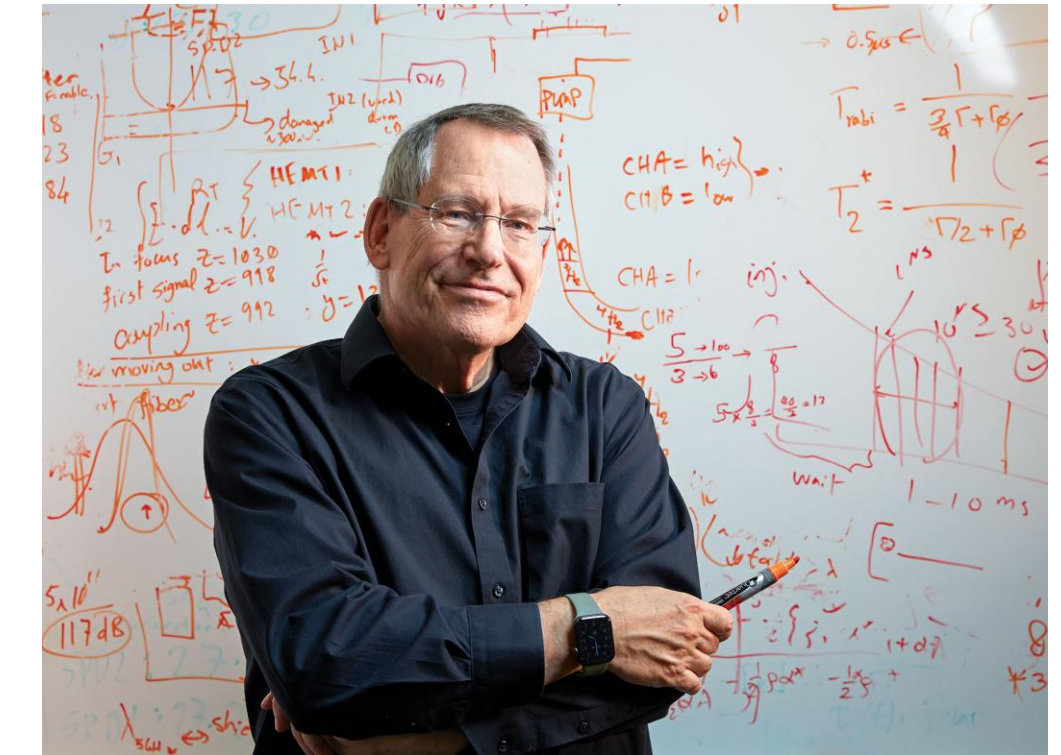
What to do with such quantum devices?

Embrace the limitations:

- Only 'short' quantum circuits
- Highly entangled quantum systems
- Frequently hybrid architecture



- Sampling problems
 - Optimization
- Quantum simulation
- Quantum Machine Learning



Preskill, John. "Quantum computing in the NISQ era and beyond." *Quantum* (2018)

Given a quantum circuit, sample from its output probability distribution.

Find a probability distribution:

- classically intractable to simulate
- tractable on quantum noisy quantum computer

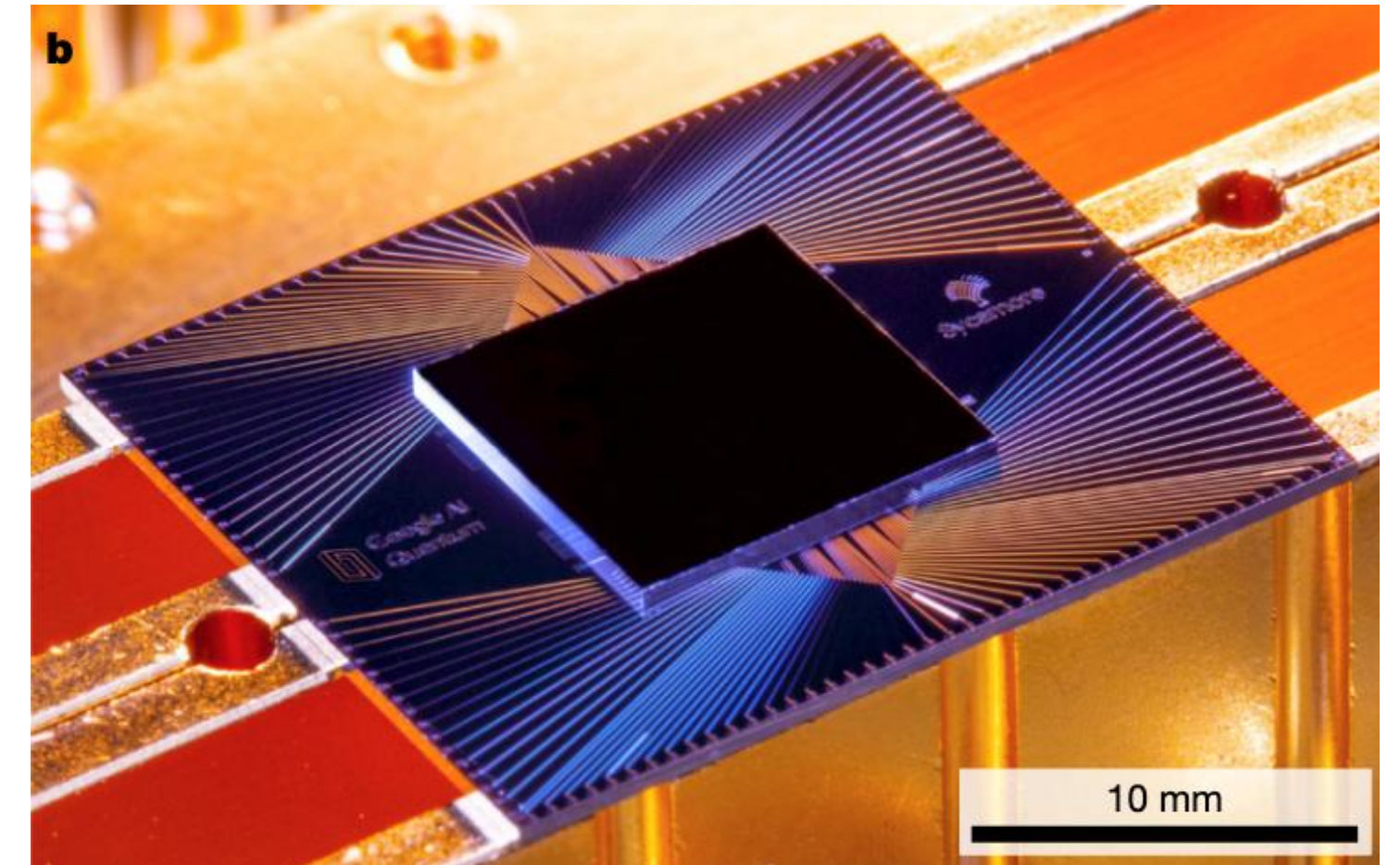
Boson Sampling
Gaussian Boson Sampling
Random Circuit Sampling



Collect samples and
Claim advantage!

Drawbacks?

- Niche problems – classical simulations improvements follow immediately
- No clear economic value



200 seconds quantum vs ~10000 years classical

Arute, Frank, et al. "Quantum supremacy using a programmable superconducting processor." *Nature* (2019).

- Field at the intersection of Machine Learning and Quantum Computing
- Fault-tolerant: full quantum routines
- NISQ: hybrid architectures

(Q)ML is NISQ friendly:

- Many repetitions of similar computation
- ↓
- Many shallow circuit runs

Main approaches:

- Quantum Kernel Methods
- Quantum Neural Networks

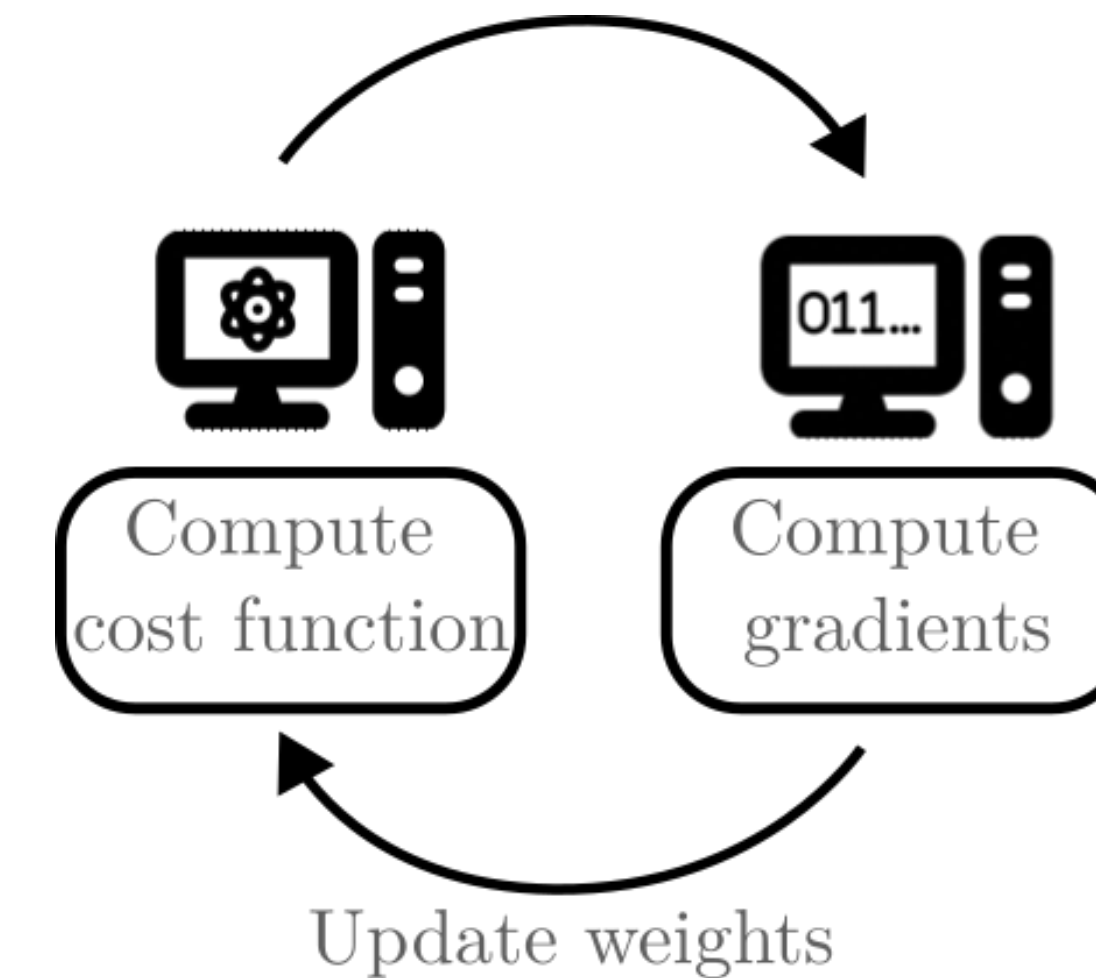
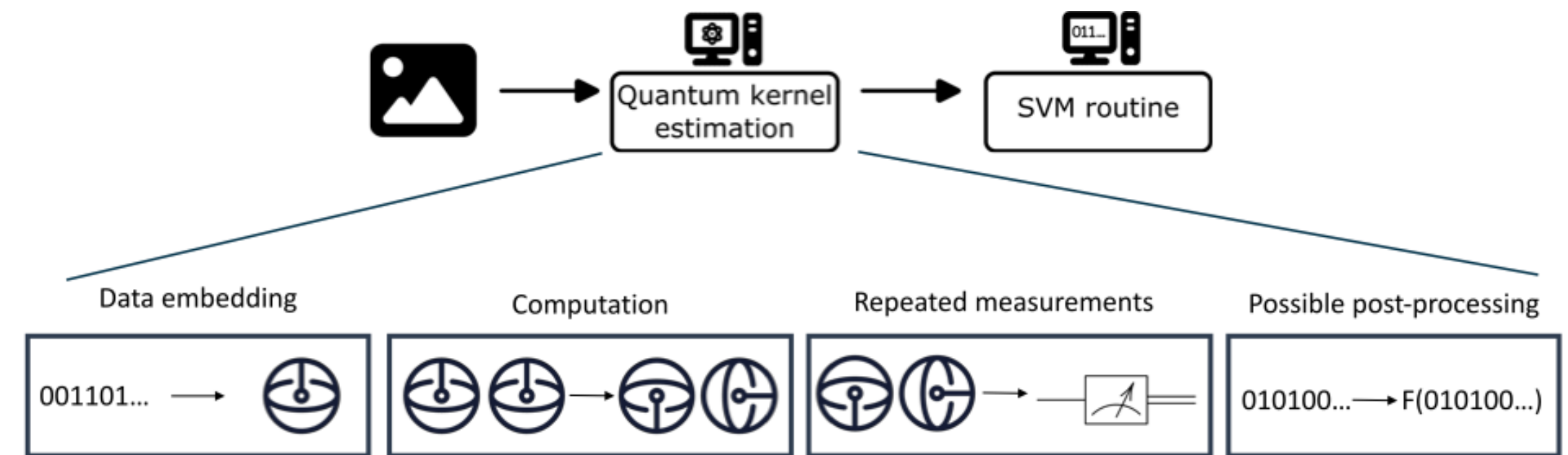
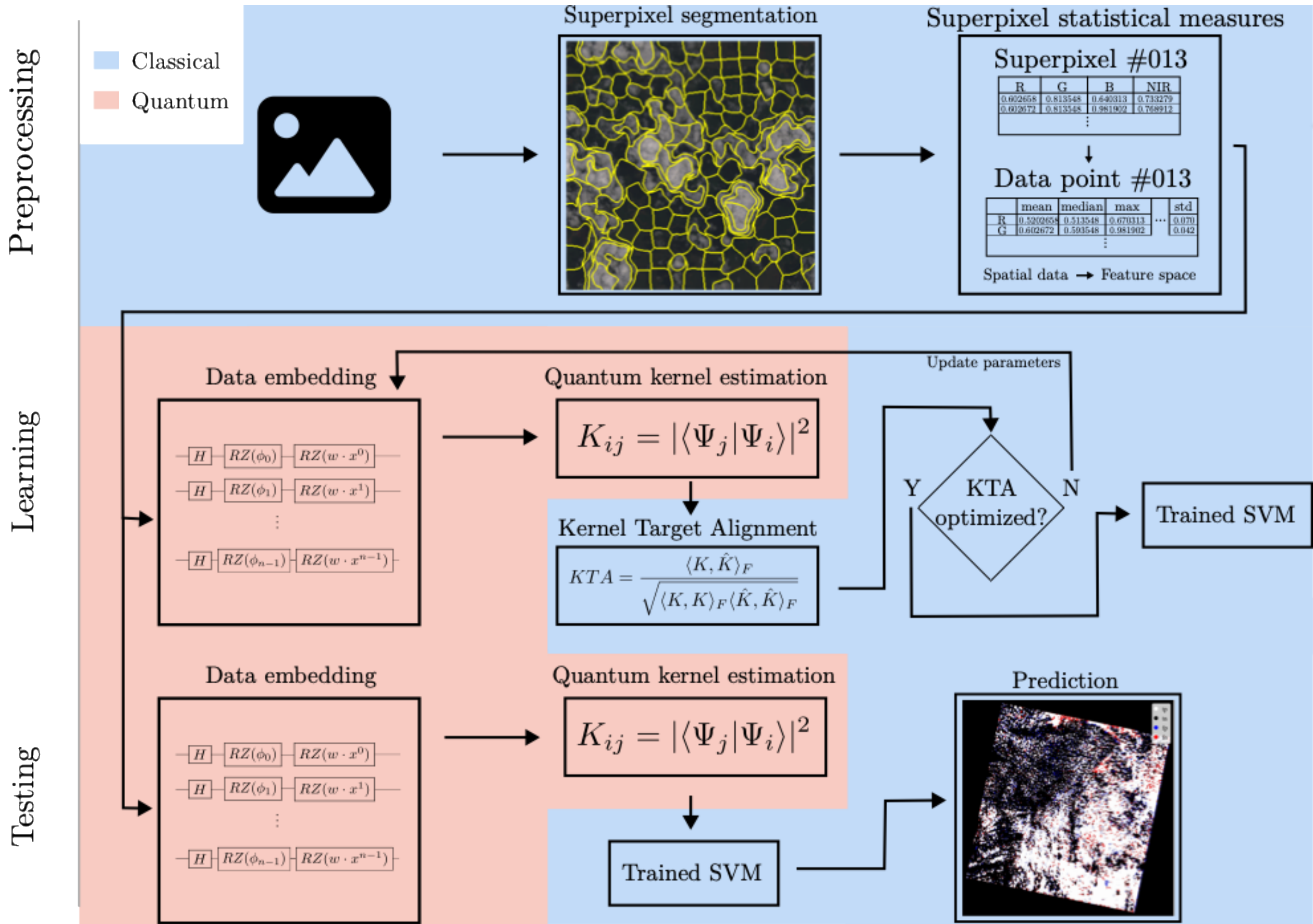


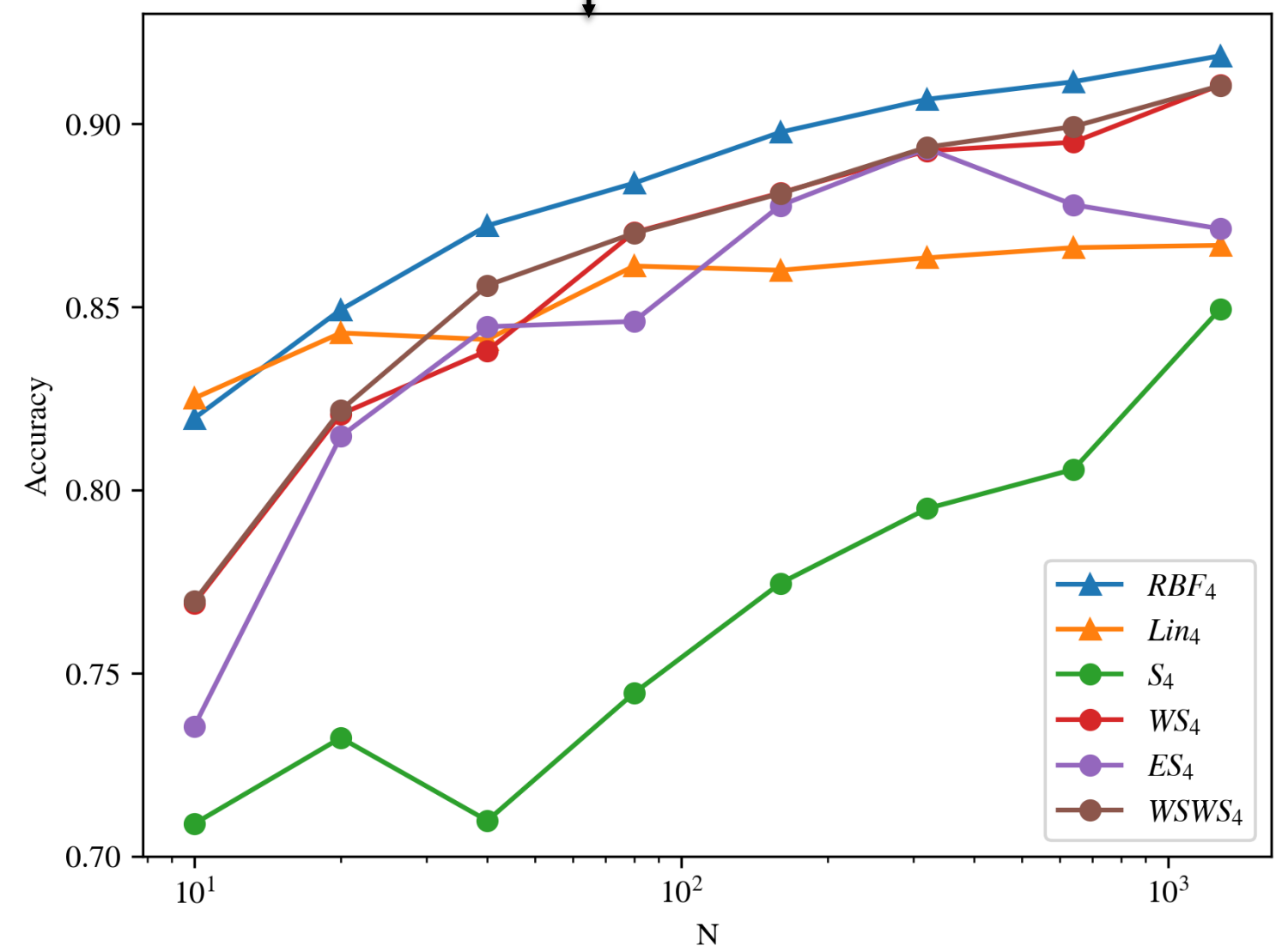
TABLE 1. THE SUMMARY OF SELECTED STUDIES APPLYING QML TO REMOTE SENSING TASKS.

STUDY/REF.	DATASET/TASK	ML PROBLEM	QML METHOD	QML ACC.	CLASS. ACC.	SOTA	MODE	NOTES
Chang et al. [106]	EuroSAT, SAT4/land cover	Binary classification	QNN	98%	—	98.5% [107]		—
Sebastianelli et al. [92]	EuroSAT/land cover	Multiclass classification	QNN	92%	83%	98.5% [107]		—
Sebastianelli et al. [108]	Volcanic eruption prediction	Binary classification	QNN	96%	88%	85%		—
Mauro et al. [109]	ΦSat-2/water quality monitoring	Regression	QNN	RMSE 0.198	RMSE 0.214	—		Advantage in the decrease of model parameters claim
Rodriguez-Grasa et al. [110]	Photovoltaic array [111]	Binary classification	QK	86.5(2.5)%	89.4(2.5)%	—		—
Wijata et al. [95]	HYPERVIEW/bare soil detection	Binary classification	QK	88.2(3.5)%	90.4(3.4)%	—		—
Gupta et al. [112]	Artificial/land cover	Binary classification	QK	≈ 80%	≈ 50%	—		Advantage for QK-aligned dataset
Miroszewski et al. [20]	38-Clouds/cloud detection	Binary classification	QK	91.1(3.1)%	91.9(1)%	96% [113]		—
Otgonbaatar and Datcu [69]	Indian Pines/land cover	Binary classification	QA	92%	90%	—		Multiple binary comparisons, on par
Otgonbaatar and Datcu [69]	PolSAR/land cover	Binary classification	QA	99%	99%	—		—
Delilbasic et al. [19]	SemCity Toulouse/land cover	Binary classification	QA	87.4%	87.6%	—		—
Delilbasic et al. [67]	SemCity Toulouse/land cover	Multiclass classification	QA	86.6%	86%	—		Advantage in computational time scaling claim
Delilbasic et al. [67]	ISPRS Postdam/land cover	Multiclass classification	QA	81.2%	85.2%	—		Advantage in computational time scaling claim
Pasetto et al. [68]	SeaBAM/chlorophyll concentration	Regression	QA	MSE 10.88	MSE 8.4	—		—

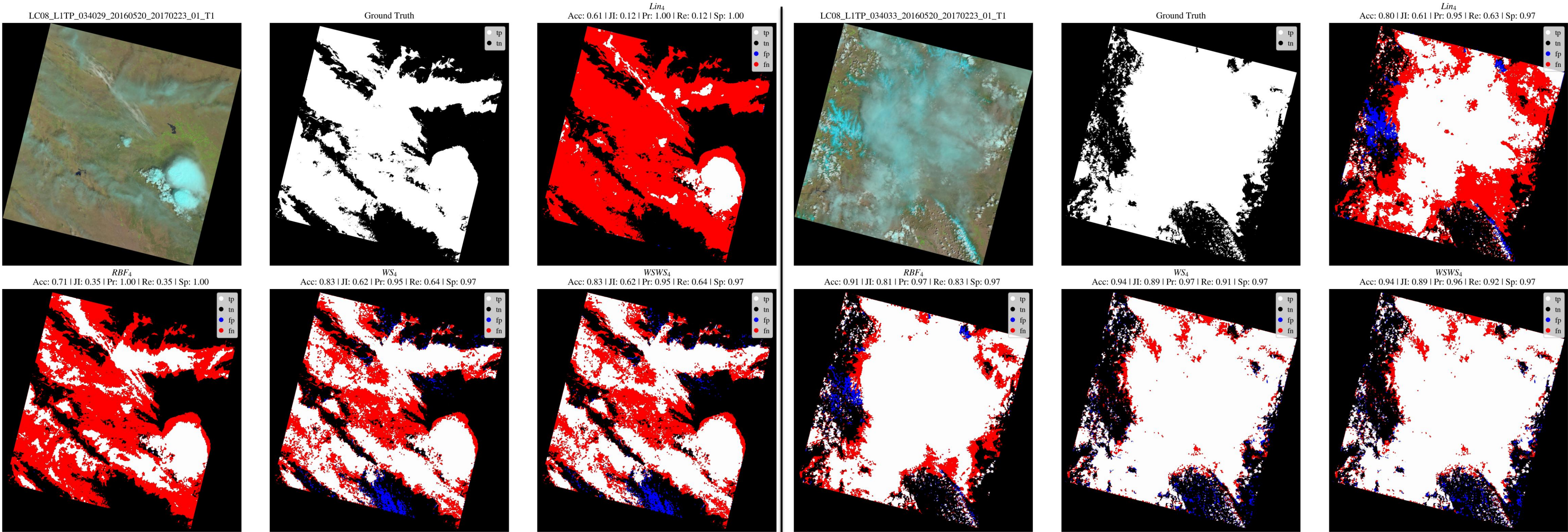
Miroszewski, Artur, et al. "Quo Vadis, Quantum Machine Learning?: Quantum kernel methods meet Earth observation." *IEEE GRSM*(2026).



Miroszewski, Artur, et al. "Detecting clouds in multispectral satellite images using quantum-kernel support vector machines." *IEEE JSTARS* 16 (2023)



Quantum Machine Learning in Remote Sensing



Miroszewski, Artur, et al. "Detecting clouds in multispectral satellite images using quantum-kernel support vector machines." *IEEE JSTARS* 16 (2023)

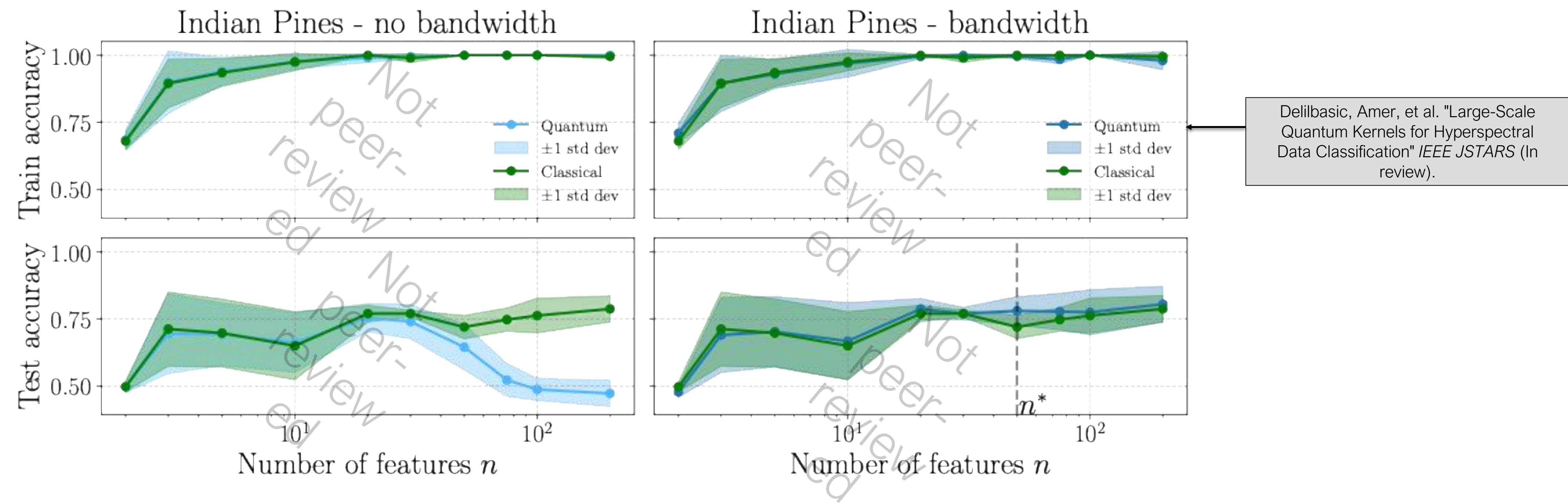
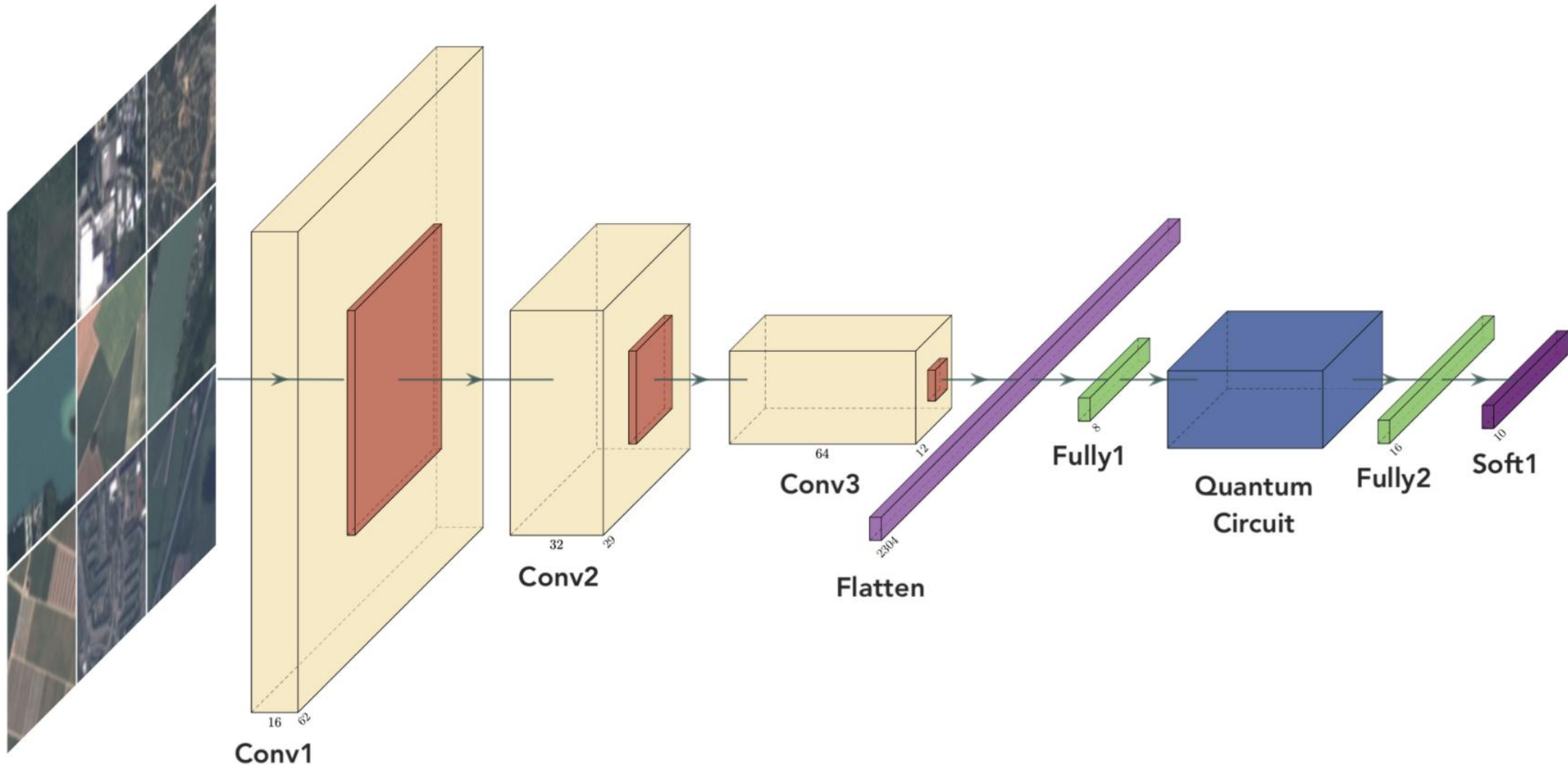


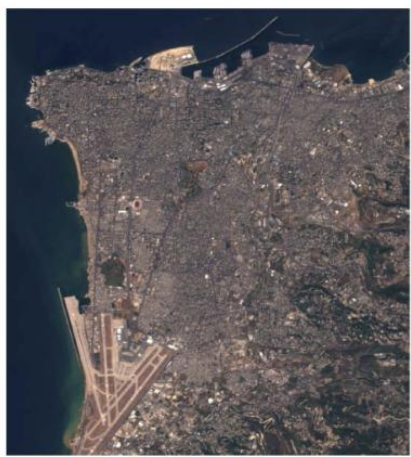
TABLE III: Indian Pines—multiclass classification performance metrics of different state-of-the-art models and the presented quantum kernel SVM. The models are trained on 4 splits containing 100-100-200 train-validation-test data points from classes *corn-mintill* (class 3), *soybean-notill* (class 10), *hay-windrowed* (class 8) and *grass-pasture* (class 5). Results are for number of features $n = 50$, reported as mean $\pm$ standard deviation over 4 splits. Best results for each metric are highlighted in bold.

Model	Accuracy	Precision (weighted)	Recall (weighted)	Cohen's kappa	F1 (weighted)
AdaBoost [68]	0.574 $\pm$ 0.113	0.603 $\pm$ 0.141	0.574 $\pm$ 0.113	0.412 $\pm$ 0.147	0.556 $\pm$ 0.142
Decision Tree [69]	0.657 $\pm$ 0.079	0.698 $\pm$ 0.099	0.657 $\pm$ 0.079	0.523 $\pm$ 0.107	0.654 $\pm$ 0.078
Gaussian Naive Bayes [70]	0.581 $\pm$ 0.072	0.627 $\pm$ 0.064	0.581 $\pm$ 0.072	0.414 $\pm$ 0.111	0.579 $\pm$ 0.088
GentleBoost [71]	0.721 $\pm$ 0.049	0.749 $\pm$ 0.051	0.721 $\pm$ 0.049	0.612 $\pm$ 0.064	0.723 $\pm$ 0.048
K-Nearest Neighbors [72]	0.815 $\pm$ 0.034	0.829 $\pm$ 0.031	0.815 $\pm$ 0.034	0.744 $\pm$ 0.045	0.813 $\pm$ 0.035
Logistic Regression [73]	0.740 $\pm$ 0.090	0.755 $\pm$ 0.090	0.740 $\pm$ 0.090	0.638 $\pm$ 0.124	0.737 $\pm$ 0.093
Random Forest [74]	0.723 $\pm$ 0.060	0.752 $\pm$ 0.065	0.723 $\pm$ 0.060	0.615 $\pm$ 0.088	0.721 $\pm$ 0.063
RUSBoost [75]	0.516 $\pm$ 0.032	0.463 $\pm$ 0.055	0.516 $\pm$ 0.032	0.343 $\pm$ 0.045	0.435 $\pm$ 0.018
SVM _C [76]	0.826 $\pm$ 0.071	0.854 $\pm$ 0.042	0.826 $\pm$ 0.071	0.759 $\pm$ 0.099	0.825 $\pm$ 0.072
SVM _Q [7]	0.833 $\pm$ 0.031	0.845 $\pm$ 0.032	0.833 $\pm$ 0.031	0.768 $\pm$ 0.042	0.830 $\pm$ 0.033

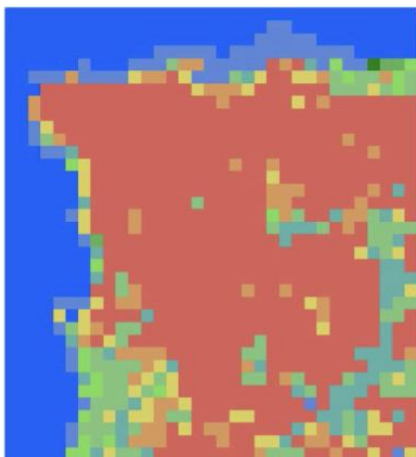
Quantum Machine Learning in Remote Sensing



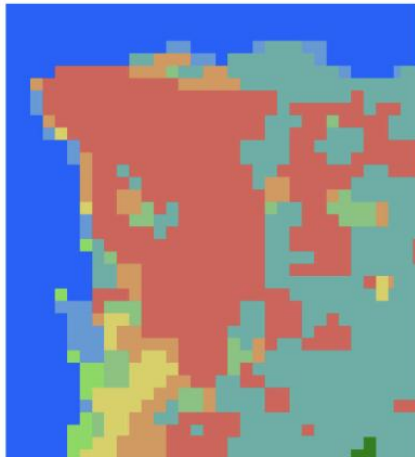
- | | | | | |
|-------------|------------|------------|-------------|----------|
| Annual Crop | Herb. Veg. | Industrial | Perm. Crop | River |
| Forest | Highway | Pasture | Residential | Sea&Lake |



Sentinel-2 image [79]



coarse-to-fine quantum classifier



Wide-ResNet [80]



JEM [80]

Sebastianelli, Alessandro, et al. "On circuit-based hybrid quantum neural networks for remote sensing imagery classification." *IEEE JSTARS* (2021)

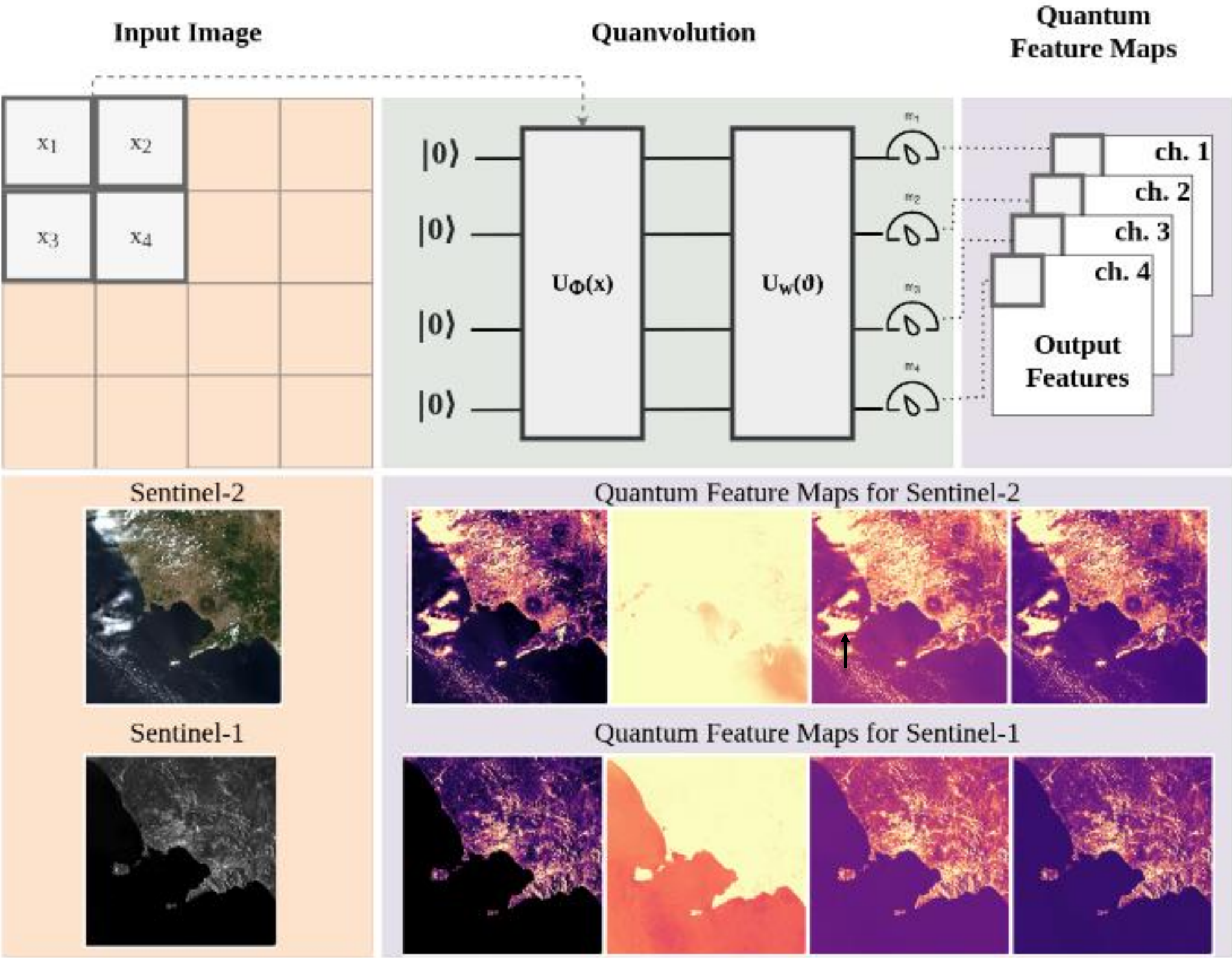
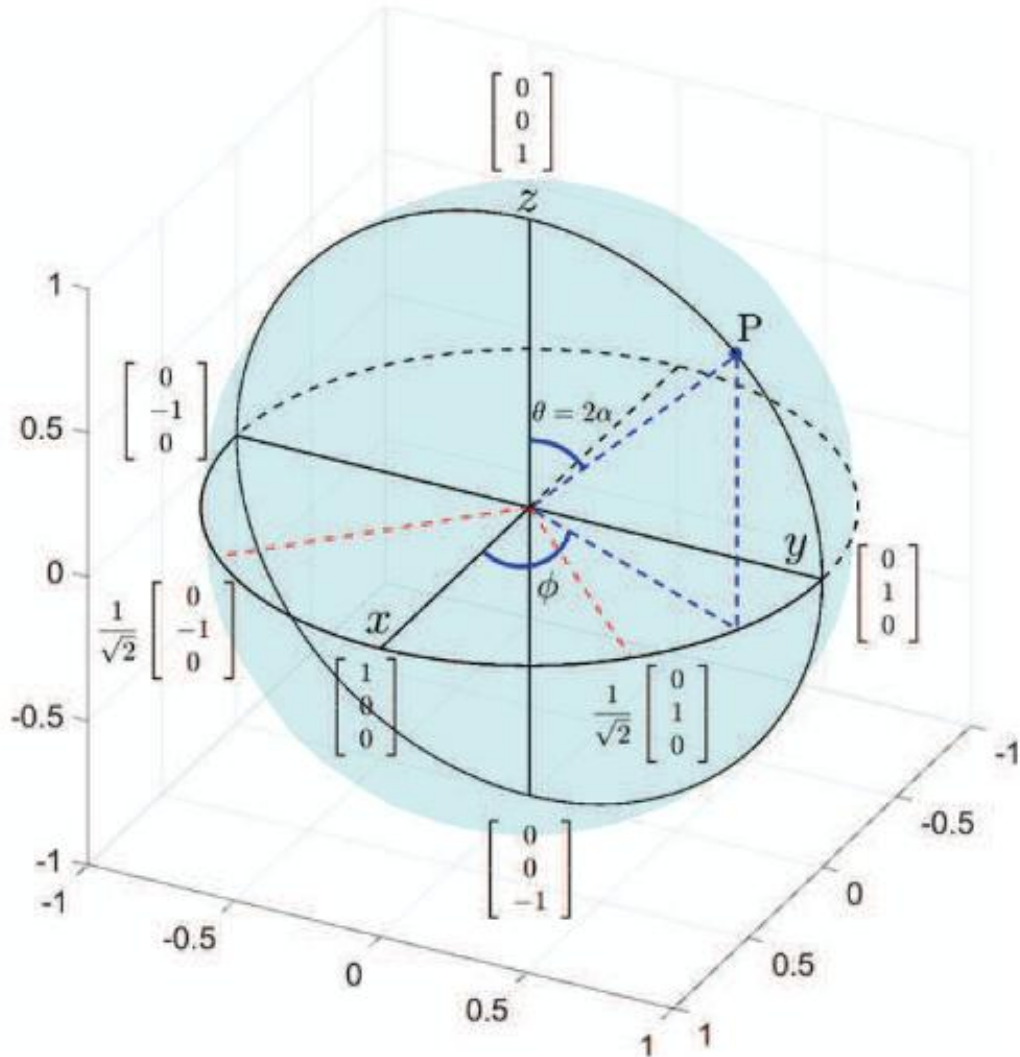


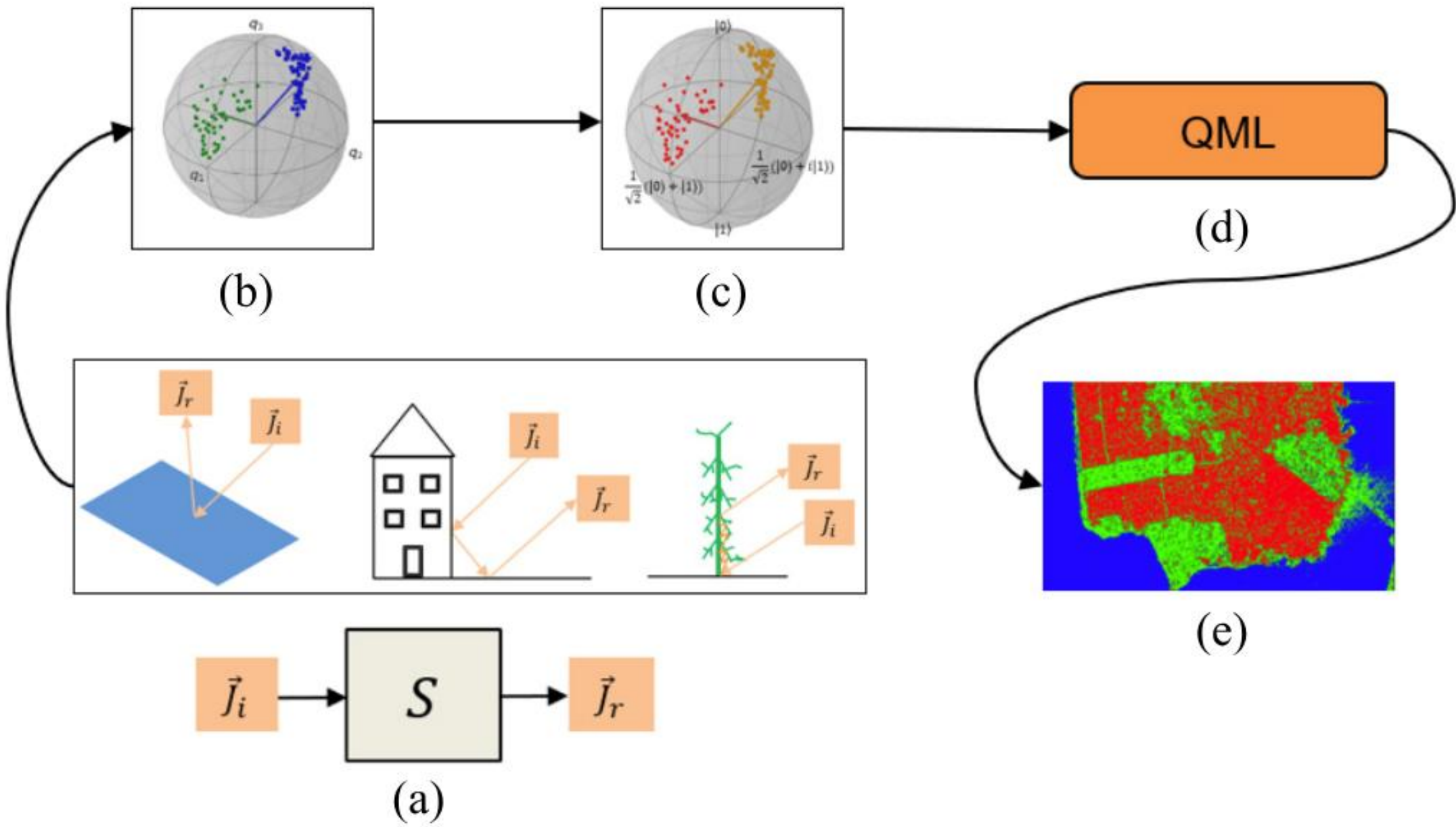
TABLE IV
RESULTS OF BIRCH CLUSTERING FOR QCNN AND CNN FEATURE MAPS.

Class	CNN	QCNN
Annual Crop	0.87	0.99
Forest	0.85	0.89
H. Vegetation	0.99	0.99
Highway	0.97	0.99
Industrial	0.99	0.99
Pasture	0.85	0.92
P. Crop	0.99	0.93
Residential	0.90	0.94
River	0.97	0.95
Sea Lake	0.99	0.99
Mean Acc.	0.94	0.96

Sebastianelli, Alessandro, et al. "Quanv4eo: empowering earth observation by means of quanvolutional neural networks." *IEEE TGRS* (2025).



Bhattacharya, Avik, and Abhinav Verma.
"Bloch Sphere Representation of Polarimetric
SAR Targets." *IEEE GRSL* (2025).



Otgonbaatar, Soronzonbold, and Mihai Datcu.
"Natural embedding of the stokes parameters of
polarimetric synthetic aperture radar images in a gate-
based quantum computer." *IEEE TGRS* (2021).

1. Physics/Engineering

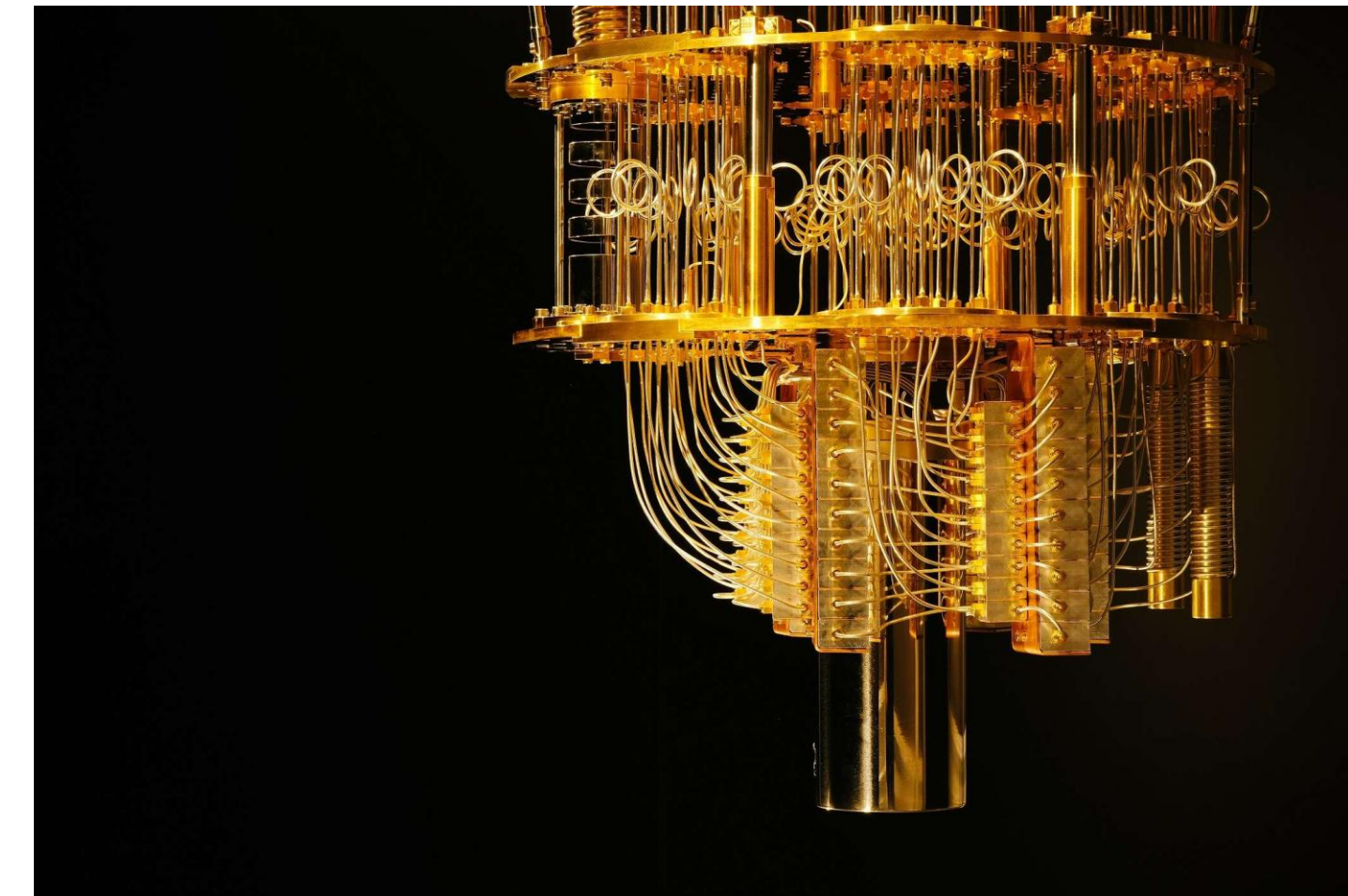
- Qubit fragility, high error rates
- Scalability, connectivity

2. Algorithms

- NISQ
 - Little or no quantum advantage
 - Not very useful problems
 - Excellent optimization of classical computers
 - Benchmarking
- Fault-tolerant
 - Mainly Shor's and Grover's algorithms
 - QEC not ready

3. Cost & Accessibility

4. Workforce



Kim, Youngseok, et al.
"Evidence for the utility of quantum computing before fault tolerance."
Nature 618.7965 (2023): 500-505.

Anonymous,
"Quantum Disadvantage
Or, simulating IBM's
'quantum utility'
experiment with a
Commodore 64"
SIGBOVIK 2024



Technical Committees to promote the knowledge and application of high-performance and innovative computing methods in geosciences and remote sensing



Schools and workshops



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QUEST-TC Webinar Series

Quantum Computing for Space Science and Technology

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Enrico Prati
University of Milan

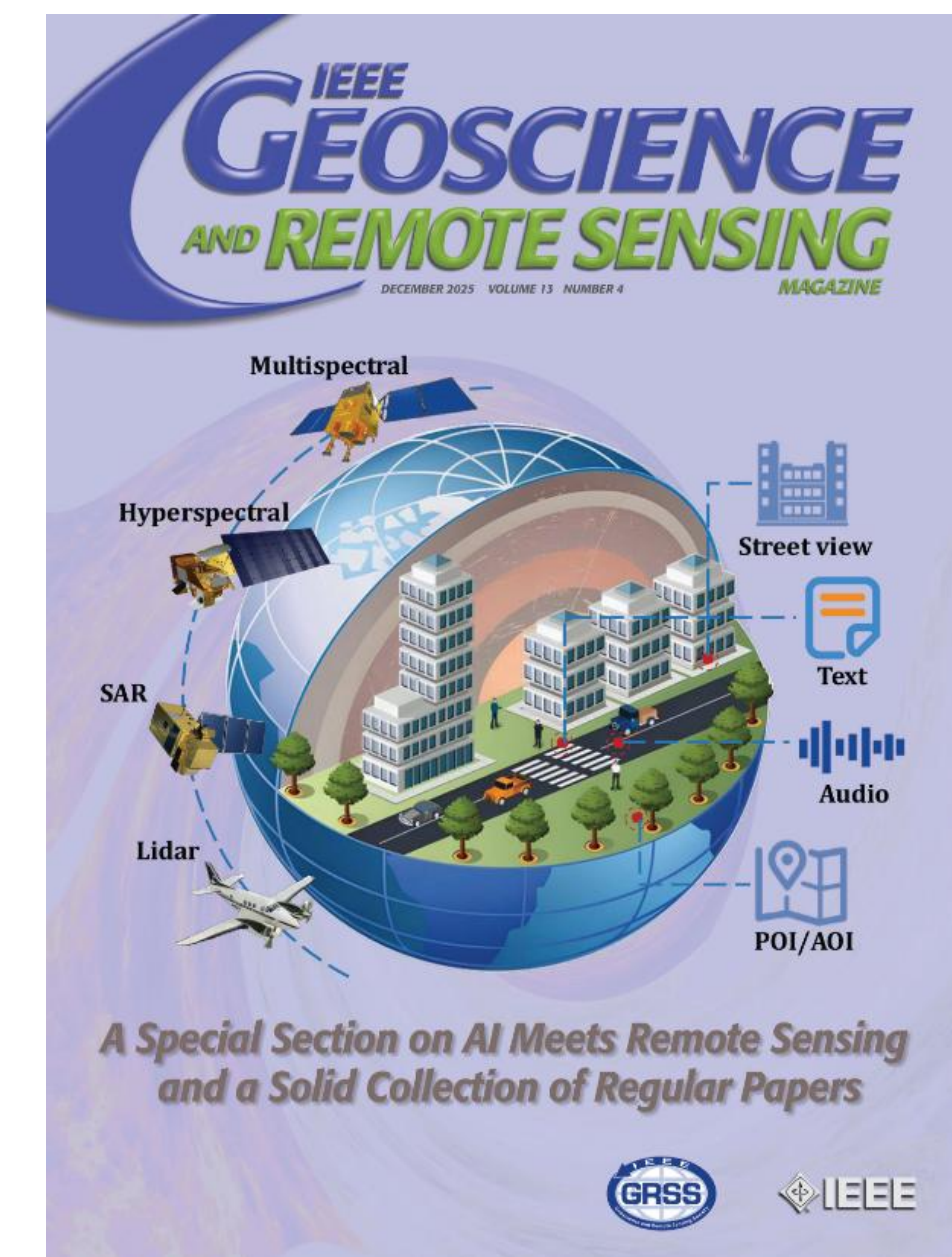
Quantum Computing for Space Science and Technology

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Dr. Ravindra Desai
Centre for Fusion, Space & Astrophysics, University of Warwick, UK

Publications

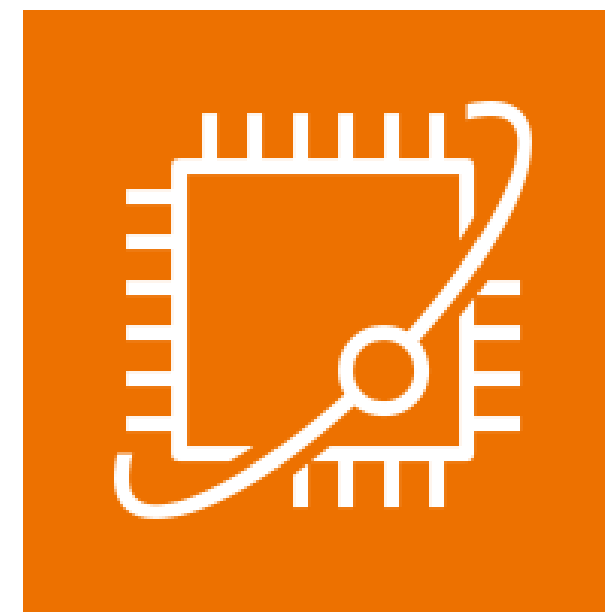


Access to software, hardware, education

Software

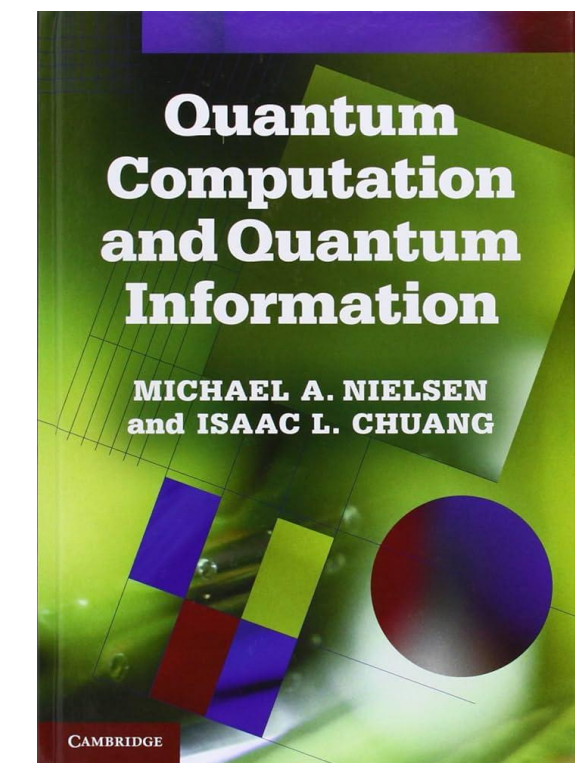
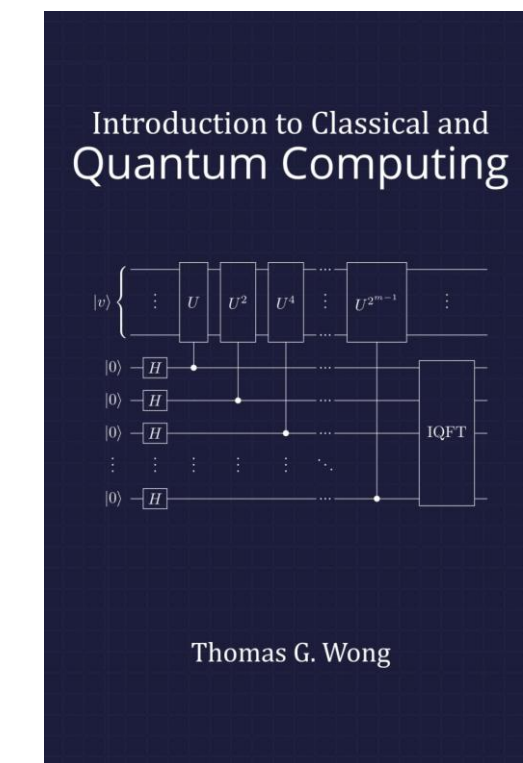


Hardware



IBM Quantum

Education



IBM Quantum Platform

Quantum hardware platforms are advancing steadily, but no one knows when quantum computers will run applications that broadly benefit society.

Early applications of quantum computing will be primarily scientific. Broader economic impact will eventually follow.

Quantum random circuit sampling experiments demonstrate today's NISQ quantum processors can perform some tasks that are beyond the reach of today's classical supercomputers; this is a notable milestone, but not of great practical interest.

For decades, experimental progress on QEC lagged far behind theory, but this is starting to change.

Early fault-tolerant machines, capable of reliably executing one million or more quantum operations, will be able to perform some tasks that are beyond the reach of classical devices.

We do not know yet which quantum computing modalities will be best suited for scaling to large systems that solve hard problems.

Eisert, Jens, and John Preskill. "Mind the gaps: The fraught road to quantum advantage." *arXiv preprint arXiv:2510.19928* (2025).

Thank you for your attention. Questions ?

Artur.Miroszewski@esa.int



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